

DC Link Capacitor

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1 Introduction

DC-link capacitors are among the most critical and failure-prone components in modern power electronic converters. Field data consistently show that capacitor degradation is a major cause of inverter downtime in photovoltaic systems, wind turbines and industrial drives. Unlike IGBTs and diodes whose dominant failure modes arise from thermo-mechanical fatigue capacitors exhibit deterministic wear out driven by electrolyte evaporation, dielectric degradation, rising equivalent series resistance (ESR) and thermally accelerated chemical aging. These mechanisms strongly depend on operating temperature, voltage stress, ripple current and ambient environment, causing predictable lifetime reduction over long-term operation. In converter-dominated grids, DC-link capacitors experience a combination of high-frequency switching ripple, low-frequency load cycling and elevated enclosure temperatures. Manufacturers report that capacitor lifetime typically halves for every 7–10°C increase in core or hotspot temperature, and voltage derating is often required to meet the 10–20 year service life expected in renewable-energy power converters. These sensitivities make capacitor reliability a primary bottleneck in ensuring long-term availability and grid-support capability of inverter-based resources.

1.1 Identified Research Gap

Despite extensive literature on capacitor failure mechanisms, practical tools for capacitor lifetime estimation remain fragmented. Electrolytic capacitors use a variety of empirical manufacturer-specific models, while film capacitors employ different voltage- and temperature-acceleration laws. Generic models such as Arrhenius voltage–temperature acceleration are widely used but rarely integrated into complete electro-thermal simulation pipelines. Existing commercial tools provide only vendor-specific calculators, and high-fidelity FEM approaches are computationally unsuitable for mission-profile simulations.

1.2 Repository Contribution

This documentation presents the capacitor-lifetime module of PEARL, an open-source electro-thermal reliability framework for power converters. The module implements:

- brings together multiple lifetime models for both electrolytic and film capacitors in one model,
- connect electrical stress, loss calculation and thermal modeling to lifetime prediction,
- evaluate long-duration mission profiles, and
- support reproducible research and practical field applications.

These models are fully integrated with PEARL’s electrical and thermal domains, enabling per-sample lifetime estimation under arbitrary operating profiles. Using Miner’s rule for damage accumulation, the framework computes the equivalent lifetime and remaining useful life (RUL) of DC-link capacitors under realistic converter mission profiles. This represents the first open-source, physics-grounded and extensible modeling environment aimed at long-term capacitor reliability assessment in power electronic converters.

2 Methodology

This section describes the modelling workflow developed to estimate the lifetime of DC-link capacitors. The framework follows a physics-informed, multi-domain approach that connects electrical loading, thermal behaviour and lifetime degradation through standardized capacitor lifetime models and datasheet-derived curves.

2.1 Electrical Stress Modelling

The process begins by computing the electrical stresses imposed on each capacitor in the DC-link. For every time step of the mission profile:

- the per-capacitor DC voltage V_{cap} is evaluated,
- the RMS ripple current I_{rms} is obtained from converter operating conditions,
- the leakage current is estimated using an insulation-resistance model.

These electrical quantities serve as inputs to both the loss model and thermal model.

2.2 Loss Modelling

Two primary mechanisms contribute to capacitor power dissipation:

$$P_{\text{ripple}} = I_{\text{rms}}^2 \cdot \text{ESR}_{\text{eff}}, \quad (1)$$

$$P_{\text{leak}} = \frac{V_{\text{cap}}^2}{R_{\text{leak}}}. \quad (2)$$

The total loss is therefore

$$P_{\text{loss}} = P_{\text{ripple}} + P_{\text{leak}}. \quad (3)$$

The effective ESR may be temperature- or frequency-dependent when such information is available.

2.3 Thermal Modelling and Calibration

A single-node thermal model is employed to compute the capacitor core temperature:

$$T_{\text{core}} = T_{\text{amb}} + k R_{\text{th}} P_{\text{loss}}, \quad (4)$$

where R_{th} is the thermal resistance and k is a calibration factor. Calibration factor ensures that the thermal model reproduces manufacturer-rated temperature behaviour. Datasheet-provided voltage-derated maximum permissible temperatures are also enforced to guarantee that each capacitor remains within safe operating limits.

2.4 Lifetime Modelling

To compute the capacitor lifetime, several models are supported, including manufacturer-specific and generic forms:

- Nichicon, Rubycon, Panasonic (electrolytic capacitors),
- Cornell-Dubilier, Faratronic (film capacitors),
- a generic Arrhenius voltage–temperature acceleration model,
- a graph-based model derived directly from datasheet lifetime curves.

Each model provides a predicted time-to-failure L_i corresponding to the operating conditions at time step i .

2.5 Cumulative Damage Assessment (Miner’s Rule)

For dynamic mission profiles, lifetime consumption is accumulated using Miner’s rule:

$$d_i = \frac{\Delta t_i}{L_i}, \quad (5)$$

$$D = \sum_i d_i, \quad (6)$$

where Δt_i is the time spent at condition i and L_i is the predicted lifetime at that condition. The equivalent lifetime under the mission profile is obtained as

$$L_{eq} = \frac{T_{profile}}{D}. \quad (7)$$

2.6 Limit Checking and Safety Constraints

To ensure safe operation, the model verifies:

- maximum allowable capacitor core temperature,
- maximum ripple current ratings,
- maximum allowed DC voltage and surge limits.

Violations of these constraints trigger warnings or errors within the tool. Overall, the methodology transforms mission-profile data into lifetime predictions by linking electrical stress, losses, thermal behaviour and degradation using validated physics-based and datasheet-derived models.

3 Model architecture

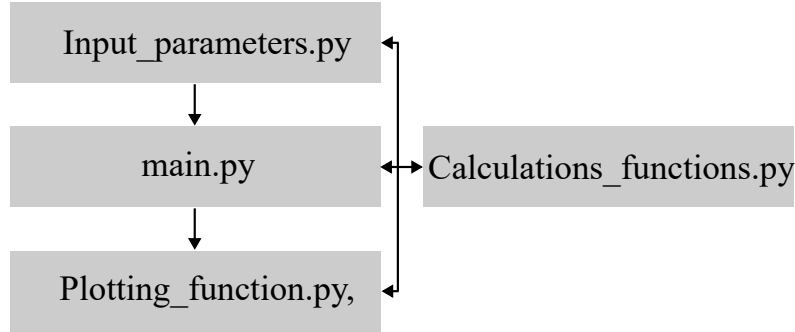


Figure 1: Model architecture.

The complete computational framework of the developed model is illustrated in Fig. 1. The flowchart summarizes the interaction among the main Python modules used for DC-link capacitor lifetime estimation. The process begins with the definition of the capacitor electrical and thermal parameters, together with the environmental conditions and inverter operating parameters. Based on these inputs, the electrical stress on the DC-link capacitor is evaluated by computing the RMS ripple current and the corresponding power losses. The calculated electrical losses are then used within a thermal model to estimate the capacitor core temperature. Subsequently, the capacitor lifetime is determined using temperature- and voltage-dependent lifetime models, combined with a cumulative damage approach based on Miner's rule to account for the mission profile. Finally, the computed electrical, thermal, and lifetime metrics are post-processed and visualized. Each block in the flowchart corresponds directly to a dedicated functional module within the Python codebase.

Detailed description of python modules

This section provides a comprehensive overview of the Python modules that constitute the developed DC-link capacitor lifetime assessment framework. Each module in the is designed to fulfill a specific role, ranging from input parameter definition and mission profile specification to electrical stress and loss computation, thermal modelling, lifetime estimation, and result visualization. Together, these modules form a structured and modular workflow, enabling clear data flow and seamless interaction between computational stages. The following subsections describe each module in detail, outlining their primary objectives, key functions, exchanged variables, and their interaction with other components of the framework.

3.1 Input_parameters.py

This module initializes and stores all input parameters required by the model. It defines the operating conditions, mission profile, thermal and electrical device parameters, and lifetime constants.

3.1.1 Inverter configuration parameters

Table 1 variables used in the inverter simulation. In principle, a detailed inverter simulation is not required for the lifetime model. The electrical quantities required for the DC-link capacitor analysis are the inverter phase current I_s , modulation index M , and phase angle ϕ , which are used to compute the capacitor RMS ripple current. Alternatively, the ripple current can be provided directly as an input, bypassing the need for inverter-related calculations. The function `compute_power_flow` in `Input_parameters.py` serves solely as a convenience layer, converting active and reactive power mission profiles (P , Q) and basic inverter parameters into the required electrical inputs.

Table 1: Electrical input and inverter configuration parameters used in the electro-thermal simulation.

Variable	Type	Unit	Description
P	array	W	Active power profile at the inverter AC side (mission-profile input).
Q	array	VA	Reactive power profile at the inverter AC side (mission-profile input). Positive for capacitive and negative for inductive
Vs	array	V	Phase RMS AC-side voltage of the inverter (line-to-neutral).
V_dc	array	V	DC-link voltage applied to the inverter. Also used to check V_{CE} limits.
M	float	–	Inverter modulation index. Here fixed to $M = 1$ (no overmodulation modelled).
modulation_scheme	str	–	PWM strategy for a three-phase inverter: "svm" (space vector / 3rd-harmonic injection) or "spwm" (sinusoidal PWM).
Is	array	A	Phase RMS current on the AC side of the inverter (output current per phase).
phi	array	rad	Phase angle between phase voltage and current. Positive for capacitive, negative for inductive operation.
pf	array	–	Instantaneous power factor.
S	array	VA	Apparent power profile.

3.1.2 Capacitor lifetime model parameters

These parameters described in Table 2 define the environmental conditions, electrical limits, thermal characteristics and lifetime behaviour of the DC-link capacitor.

Table 2: Capacitor, environmental, thermal and lifetime model parameters.

Variable	Type	Unit	Description
T_env	array	K	Ambient temperature profile applied to the capacitor over the mission profile.
N_series	int	–	Number of DC-link capacitors connected in series.
N_parallel	int	–	Number of DC-link capacitors connected in parallel.
capacitor_type	str	–	Type of capacitor technology used in the model ("film" or "electrolytic").
Max_voltage_datasheet_cap	float	V	Maximum allowable capacitor voltage according to the manufacturer datasheet.
Max_current_datasheet_cap	float	A	Maximum allowable RMS ripple current according to the datasheet.
Max_temperature_cap_dict	dict	K	Maximum permissible capacitor core temperature as a function of the voltage stress ratio V/V_r .
Rated_voltage_datasheet_cap	float	V	Rated capacitor voltage used as reference for voltage stress and lifetime estimation.
Rated_current_datasheet_cap	float	A	Rated RMS ripple current at nominal operating conditions.
Rated_temperature_cap	float	K	Rated core temperature corresponding to the datasheet lifetime specification.
Rated_lifetime	float	h	Manufacturer-specified capacitor lifetime at rated voltage, current, and temperature.
Thermal_resistance	float	K/W	Thermal resistance between capacitor core and ambient.
ESR_eff	float	Ω	Effective equivalent series resistance of the capacitor at the inverter switching frequency.
minimum_insulation_resistance	float	Ω	Minimum insulation resistance used to estimate leakage current and associated dielectric losses.
lifetime_graph_dictionary	dict	K,V, hours	Temperature- and voltage-dependent lifetime curves used in the graph-based lifetime interpolation model.

3.2 Calculation functions.py

The `Calculation_functions_file.py` module which has a class `Calculation_functions_class` within which we have many functions serves as the computational core of the electro–thermal and reliability modeling framework. It provides a structured collection of static methods that implement the analytical and numerical equations governing inverter operation, device losses thermal stress cycling and lifetime estimation. Below most of the important functions are explained along with the physics behind those function.

3.2.1 compute_power_flow

The function `compute_power_flow` converts a mission profile defined in terms of active and reactive power (P, Q) into the electrical quantities required by the electro–thermal model, while accounting

for the inverter configuration. Depending on the topology and modulation, the function determines the theoretical maximum RMS phase voltage $V_{s_theoretical}$ that the inverter can generate from the DC-link voltage V_{dc} and modulation index M :

$$V_{s_theoretical} = \begin{cases} \frac{M V_{dc}}{\sqrt{2}}, & \text{single-phase full-bridge,} \\ \frac{M V_{dc}}{2}, & \text{single-phase half-bridge,} \\ \frac{2\sqrt{2}}{M V_{dc}}, & \text{three-phase, svm,} \\ \frac{\sqrt{6}}{M V_{dc}}, & \text{three-phase, spwm.} \end{cases} \quad (8)$$

If no AC-side voltage profile V_s is provided, it is set equal to $V_{s_theoretical}$. Otherwise, the function enforces $V_s \leq V_{s_theoretical}$, raising an error if this condition is violated (overmodulation or inconsistent input). The power factor is computed robustly by distinguishing special cases:

$$pf = \begin{cases} 0, & P_{sw} = 0, Q_{sw} \neq 0, \\ \text{sign}(P_{sw}), & P_{sw} \neq 0, Q_{sw} = 0, \\ \text{sign}(Q_{sw}) \frac{|P_{sw}|}{S}, & P_{sw} \neq 0, Q_{sw} \neq 0, \end{cases}$$

so that purely reactive operation yields $pf = 0$, purely active operation tends to $|pf| = 1$, and the sign of pf encodes inductive ($pf < 0$) or capacitive ($pf > 0$) behaviour. The RMS current for three-phase operation S is interpreted as total three-phase apparent power and V_s as phase RMS voltage, giving

$$I_s = \frac{S}{3 V_s}.$$

The phase angle between voltage and current is computed from the power factor. For nonzero power factor ($pf \neq 0$),

$$\phi = \text{sign}(pf) \arccos(|pf|),$$

so that inductive loads ($pf < 0$) yield a negative angle and capacitive loads ($pf > 0$) a positive angle. For purely reactive operation ($pf = 0$), the function sets

$$\phi = \begin{cases} +\frac{\pi}{2}, & Q_{sw} > 0, \\ -\frac{\pi}{2}, & Q_{sw} < 0, \\ 0, & S = 0, \end{cases}$$

which covers the special case of zero apparent power.

3.2.2 `check_max_capacitor_temperature_limit`

The function `check_max_capacitor_temperature_limit` enforces the maximum allowable capacitor core temperature under voltage-dependent derating, as specified by manufacturer datasheet limits. For each operating point, the applied voltage stress is expressed as the normalized voltage ratio

$$V_{ratio} = \frac{V_{cap}}{V_{rated}}. \quad (9)$$

The corresponding maximum permissible core temperature T_{allowed} is obtained by linear interpolation of the datasheet-provided derating curve,

$$T_{\text{allowed}} = f_{\text{ds}}(V_{\text{ratio}}),$$

where f_{ds} is defined by the discrete $(V_{\text{ratio}}, T_{\text{max}})$ breakpoints stored in `Max_temperature_cap_dict`. Voltage ratios outside the datasheet range are conservatively clamped to the nearest boundary value. The function then verifies the thermal operating condition by enforcing

$$T_{\text{core}} \leq T_{\text{allowed}}.$$

If this condition is violated at any mission-profile sample, the simulation is terminated and an error is raised, indicating that the capacitor exceeds its voltage-dependent thermal limit.

3.2.3 capacitor_RMS_ripple_current

The function `capacitor_RMS_ripple_current` computes the RMS ripple current flowing through the DC-link capacitor using an analytical expression derived for PWM voltage-source inverters. The formulation is based on the model introduced by Kolar *et al.* [1] and expresses the capacitor ripple current as a function of the inverter phase current, modulation index, and displacement angle. For a three-phase inverter, the DC-link capacitor RMS ripple current is given by

$$I_{\text{dcl}} = \sqrt{2} I_s \sqrt{M \left(\frac{\sqrt{3}}{4\pi} + \left(\frac{\sqrt{3}}{\pi} - \frac{9M}{16} \right) \cos^2 \phi \right)}, \quad (10)$$

where I_s is the RMS phase current, M is the modulation index, and ϕ is the phase angle between the inverter output voltage and current. This expression captures the combined influence of modulation depth and load power factor on the DC-link capacitor stress and provides the electrical input required for subsequent loss, thermal, and lifetime calculations.

3.2.4 core_temperature_calculationsI_cap

The function `core_temperature_calculationsI_cap` estimates the capacitor core temperature using a calibrated lumped electro-thermal model, following the approach presented in [2]. The model accounts for both ripple-current-induced conduction losses and dielectric leakage losses within the capacitor. The ripple losses are computed as

$$P_{\text{ripple}} = I_{\text{cap}}^2 \text{ESR}_{\text{eff}}, \quad (11)$$

where I_{cap} is the RMS ripple current per capacitor and ESR_{eff} is the effective equivalent series resistance at the inverter switching frequency. In addition, dielectric leakage losses are estimated from the insulation resistance as

$$P_{\text{leak}} = \frac{V_{\text{cap}}^2}{R_{\text{ins}}}, \quad (12)$$

with V_{cap} denoting the per-capacitor DC voltage and R_{ins} the minimum insulation resistance. The total internal power dissipation is therefore

$$P_{\text{loss}} = P_{\text{ripple}} + P_{\text{leak}}. \quad (13)$$

The capacitor core temperature is then obtained using a lumped thermal resistance model,

$$T_{\text{core}} = T_{\text{env}} + k R_{\text{th}} P_{\text{loss}}, \quad (14)$$

where T_{env} is the ambient temperature, R_{th} is the capacitor thermal resistance, and k is a calibration factor identified such that the model reproduces the rated core temperature under datasheet operating conditions [2].

3.2.5 core_temperature_calibration_factor

The function `core_temperature_calibration_factor` identifies the calibration constant used in the lumped electro-thermal capacitor model. This factor accounts for unmodelled thermal effects and ensures consistency between the analytical temperature model and the manufacturer's rated operating conditions. At the rated operating point, the internal capacitor losses are computed as the sum of ripple-current conduction losses and dielectric leakage losses,

$$P_{\text{loss}} = I_{\text{cap}}^2 \text{ESR}_{\text{eff}} + \frac{V_{\text{cap}}^2}{R_{\text{ins}}}. \quad (15)$$

The calibration factor k is then obtained by enforcing that the thermal model reproduces the rated core temperature,

$$k = \frac{T_{\text{core,r}} - T_{\text{env,r}}}{R_{\text{th}} P_{\text{loss}}}, \quad (16)$$

where $T_{\text{core,r}}$ and $T_{\text{env,r}}$ denote the rated core and ambient temperatures, respectively, and R_{th} is the capacitor thermal resistance. Once identified, the same calibration factor is used in the core temperature estimation for all mission-profile operating points, thereby anchoring the thermal model to datasheet conditions.

In [2], a constant correction factor of 1.5 is adopted for aluminum electrolytic capacitors, following the refinement proposed by Cornell Dubilier to account for the increase in ESR and associated temperature rise over the capacitor lifetime. This factor effectively penalizes ripple-current-induced heating more strongly than ambient temperature, reflecting technology- and manufacturer-specific ageing behaviour. In the present work, a generalized capacitor lifetime framework is developed to support different capacitor technologies and manufacturers. As a result, a fixed correction factor cannot be universally assumed. Instead, the calibration factor is identified directly from rated datasheet conditions, ensuring that the electro-thermal model reproduces the specified core temperature at nominal operating points. While an idealized value of this factor would be unity for a perfectly modelled system, deviations from unity naturally arise due to unmodelled thermal paths, construction-specific effects, and ageing characteristics, as evidenced by the value of 1.5 reported by Cornell Dubilier for their aluminum electrolytic capacitors.

3.2.6 Nichion_lifetime_model

The function `Nichion_lifetime_model` implements the empirical lifetime model proposed by Nichicon for aluminum electrolytic capacitors, as summarized in [3]. The model captures the combined impact of operating temperature and ripple-current-induced temperature rise on capacitor lifetime. The estimated lifetime is given by

$$L = L_r 2^{\frac{T_r - T}{10}} 2^{\left(1 - \frac{\Delta T_r \left(\frac{I_r}{I}\right)^2}{K}\right)}, \quad (17)$$

where L_r is the rated lifetime at rated temperature T_r , T is the operating core temperature, ΔT_r denotes the rated temperature rise due to ripple current, I_r is the rated ripple current, I is the operating ripple current, and K is a manufacturer-specific empirical constant.

3.2.7 Rubycon_lifetime_model

The function `Rubycon_lifetime_model` implements the empirical lifetime model proposed by Rubycon for aluminum electrolytic capacitors, as reported in [3]. Unlike simpler temperature-only models, this formulation accounts for the combined effects of operating temperature, ripple-current-induced temperature rise, and voltage stress. The estimated capacitor lifetime is expressed as

$$L = L_r 2^{\frac{T_r - T}{10}} 2^{\frac{\Delta T_r - \Delta T}{10}} \left(\frac{V_r}{V} \right)^{2.5}, \quad (18)$$

where L_r is the rated lifetime at rated temperature T_r , T is the operating core temperature, ΔT_r denotes the rated ripple-induced temperature rise, and ΔT is the actual ripple-induced temperature rise. The voltage acceleration term depends on the ratio between the rated voltage V_r and the operating voltage V , with an exponent of 2.5 reflecting the dielectric stress sensitivity of aluminum electrolytic capacitors.

3.2.8 Panasonic_lifetime_model

The function `Panasonic_lifetime_model` implements the temperature-acceleration lifetime model commonly used for aluminum electrolytic capacitors, as reported in [3]. The model assumes that capacitor aging is dominated by thermal stress and follows an Arrhenius-type doubling rule, where the lifetime halves for every 10 °C increase in operating temperature. The estimated lifetime L is given by

$$L = L_r \cdot 2^{\frac{T_r - T_a}{10}}, \quad (19)$$

where L_r is the rated lifetime specified at the rated temperature T_r , and T_a is the actual capacitor core temperature.

3.2.9 Cornell_Dubilier_lifetime_model

The function `Cornell_Dubilier_lifetime_model` implements the empirical lifetime model proposed by Cornell Dubilier for metallized polypropylene film capacitors, as reported in [3]. This model accounts for the combined influence of operating temperature and voltage stress, which are the dominant ageing mechanisms for film capacitors. The estimated lifetime is expressed as

$$L = L_r 2^{\frac{T_r - T}{10}} \left(\frac{V_r F}{V} \right)^8, \quad (20)$$

where L_r is the rated lifetime specified at rated temperature T_r and rated voltage V_r , T is the operating capacitor core temperature, and V is the applied operating voltage. The coefficient F is a manufacturer-specific correction factor introduced by Cornell Dubilier to account for construction-dependent voltage stress effects.

3.2.10 Faratronic_lifetime_model

The `Faratronic_lifetime_model` estimates the lifetime of film capacitors using a voltage-acceleration only empirical model. In this formulation, the capacitor lifetime is assumed to be dominated by dielectric stress, while temperature effects are implicitly accounted for through the rated lifetime specification. The model expresses the operating lifetime L as

$$L = L_r T_r \left(\frac{V_r}{V} \right)^9,$$

where L_r is the rated lifetime at the rated voltage V_r and temperature T_r , and V is the actual operating voltage. The strong voltage exponent reflects the high sensitivity of film capacitor insulation degradation to over-voltage stress. This model is particularly suitable for film capacitors operating under relatively stable thermal conditions, as reported in [3].

3.2.11 Generic_Arrhenius_lifetime_model

The function `Generic_Arrhenius_lifetime_model` implements a generalized physics-based lifetime model that captures the combined influence of electrical stress (voltage) and thermal stress (temperature) on capacitor degradation. The model is based on an Arrhenius-type temperature acceleration combined with a power-law voltage acceleration, as summarized in [3]. The estimated lifetime L is given by

$$L = L_0 \left(\frac{V}{V_0} \right)^{-n} \exp \left(\frac{E_a}{k_B} \left(\frac{1}{T} - \frac{1}{T_0} \right) \right), \quad (21)$$

where L_0 is the reference lifetime specified at reference voltage V_0 and reference temperature T_0 . The exponent n characterizes the voltage sensitivity of the specific capacitor technology, while E_a denotes the activation energy governing thermally activated degradation mechanisms. The constant k_B is Boltzmann's constant.

3.2.12 get_lifetime_from_graph

The function `get_lifetime_from_graph` implements a data-driven lookup of capacitor lifetime from manufacturer lifetime graphs. The datasheet information is digitized and stored in the dictionary `lifetime_graph_dictionary`, where each key corresponds to a discrete temperature level and each entry contains a set of voltage stress ratios V/V_r and the associated lifetime values L (in hours). For an arbitrary operating point defined by the mission-profile core temperature T_{core} and voltage ratio $V_{\text{ratio}} = V_{\text{cap}}/V_r$, the function estimates the lifetime via a two-step interpolation procedure:

1. **Voltage interpolation:** for each available datasheet temperature T_k , the lifetime $L(T_k, V_{\text{ratio}})$ is obtained by linear interpolation along the corresponding (V_{ratio}, L) curve.
2. **Temperature interpolation:** the resulting lifetimes are then linearly interpolated between the two neighbouring temperatures that bound T_{core} , yielding $L(T_{\text{core}}, V_{\text{ratio}})$.

Operating points outside the available temperature range are conservatively clamped to the nearest datasheet temperature curve. This approach enables the lifetime evaluation of each mission-profile sample directly from datasheet-derived lifetime surfaces, ensuring that any encountered $(T_{\text{core}}, V_{\text{ratio}})$ pair is mapped to a corresponding lifetime estimate without requiring an explicit analytical ageing law.

3.2.13 miners_rule_lifetime

The function `miners_rule_lifetime` evaluates the cumulative lifetime consumption of the DC-link capacitor over a given mission profile using Miner's linear damage accumulation rule. For each mission-profile sample, the instantaneous lifetime L_i (expressed in hours) is converted into a damage increment proportional to the fraction of time spent at that operating condition. Given a sampling interval Δt , the damage contribution of the i -th sample is computed as

$$\Delta D_i = \frac{\Delta t}{L_i}, \quad (22)$$

where Δt is expressed in hours. The total accumulated damage over one complete mission profile is then

$$D = \sum_i \Delta D_i. \quad (23)$$

Failure is expected when the cumulative damage approaches unity ($D \approx 1$). The equivalent lifetime associated with a periodically repeating mission profile is finally obtained as

$$L_{eq} = \frac{T_{profile}}{D}, \quad (24)$$

where $T_{profile}$ denotes the total duration of a single mission profile. The resulting lifetime is reported in years, providing a compact metric for long-term reliability assessment under realistic operating conditions.

3.2.14 validate_lifetime_model

The function `validate_lifetime_model` enforces the consistency between the selected capacitor technology and the applied lifetime model. Since different capacitor technologies exhibit distinct ageing mechanisms, not all lifetime models are physically meaningful for every capacitor type. The function defines a set of admissible lifetime models for each supported capacitor technology (e.g., `electrolytic` and `film`). Before lifetime evaluation, the selected model name is checked against the allowed set corresponding to the specified `capacitor_type`. If either an unknown capacitor type is provided or an incompatible lifetime model is selected, the function raises an error and terminates the simulation. This validation step prevents the misuse of technology-specific lifetime laws (e.g., applying aluminum electrolytic capacitor models to film capacitors) and ensures that the lifetime assessment remains physically consistent and reproducible.

3.3 Plotting_function.py

This module provides post-processing and visualization utilities for the developed electro-thermal lifetime assessment framework. It generates a comprehensive set of time-domain and summary plots from the simulation output data, enabling visual inspection of electrical operating conditions, thermal behaviour, and resulting lifetime estimates. All deterministic simulation figures are produced within the `Plotting` function, while the `Plotting_MC` function is used to visualize the statistical lifetime distribution obtained from Monte Carlo simulations.

- **Figure 1:** Apparent power S , active power P , and reactive power Q versus time.
- **Figure 2:** Inverter AC-side RMS phase voltage V_s and RMS phase current I_s versus time.

- **Figure 3:** Power factor pf and phase angle ϕ versus time.
- **Figure 4:** DC-link voltage V_{dc} versus time.
- **Figure 5:** Ambient temperature T_{env} (in °C) versus time.
- **Figure 6:** Number of DC-link capacitors connected in series N_{series} and in parallel $N_{parallel}$ (bar plot).
- **Figure 7:** DC-link capacitor RMS ripple current I_{dcl} versus time.
- **Figure 8:** Per-capacitor DC voltage V_{cap} and per-capacitor RMS ripple current I_{cap} versus time.
- **Figure 9:** Estimated capacitor core temperature T_{core} (in °C) versus time.
- **Figure 10:** Equivalent capacitor lifetime L_{tot} (in years) obtained from Miner's rule (bar plot).
- **Figure 11:** Histogram of Monte Carlo lifetime samples with fitted Weibull probability density function (PDF), illustrating uncertainty in capacitor lifetime.

3.4 main.py

The `main.py` script serves as the top-level executable of the developed electro–thermal lifetime assessment framework. It orchestrates the complete workflow by (i) loading mission-profile and component parameters, (ii) computing the capacitor electrical stress, (iii) evaluating losses and core temperature through the calibrated thermal model, (iv) estimating lifetime using a selected lifetime law (or datasheet graph interpolation), and (v) generating and storing results for post-processing and visualization.

Parameter initialization and data import The script instantiates `Input_parameters_class` to obtain the mission-profile arrays ($P, Q, V_s, V_{dc}, I_s, \phi, pf, S$), the ambient temperature profile T_{env} , capacitor bank configuration ($N_{series}, N_{parallel}$), and all capacitor electrical/thermal/lifetime parameters (e.g., $ESR_{eff}, R_{ins}, R_{th}$, rated limits, and `lifetime_graph_dictionary`). A dedicated results directory structure is created using `create_simulation_folders`.

Electrical stress computation The DC-link capacitor RMS ripple current I_{dcl} is computed from the inverter operating point using the analytical Kolar model [1]. The per-capacitor stresses are then derived from the series/parallel configuration:

$$V_{cap} = \frac{V_{dc}}{N_{series}}, \quad I_{cap} = \frac{I_{dcl}}{N_{parallel}}.$$

The script subsequently verifies that V_{cap} and I_{cap} remain within datasheet maximum limits via `check_max_capacitor_voltage_and_current_limit`.

Electro–thermal modelling and derating validation A calibration factor k for the lumped thermal model is computed from rated conditions using `core_temperature_calibration_factor`. The calibrated model is then applied across the mission profile to estimate the capacitor core temperature T_{core} using `core_temperature_calculationsI_cap`. Finally, a voltage-dependent thermal derating constraint is enforced using `check_max_capacitor_temperature_limit`, which raises an error if T_{core} exceeds the allowable limit for the corresponding voltage ratio V_{cap}/V_r .

Lifetime model selection and mission-profile aggregation A dispatch table (`model_dispatch`) provides a unified interface to multiple lifetime models (electrolytic, film, and generic). The desired model is selected through `selected_model` and validated against the declared capacitor technology (`capacitor_type`) using `validate_lifetime_model`. In the present implementation, the graph-based interpolation model is used to compute the instantaneous lifetime L_i (hours) as a function of T_{core} and V_{cap}/V_r . The total equivalent lifetime for a repeating mission profile is then computed via Miner’s rule using `miners_rule_lifetime`, returning L_{tot} in years.

Result storage and visualization All key variables are consolidated into a pandas dataframe and stored as `df.parquet`. The first row is additionally populated with the scalar configuration quantities N_{series} , N_{parallel} and L_{tot} . The deterministic results are visualized via the plotting module by calling `Plotting(df, Figures_dir)`.

Monte Carlo uncertainty analysis To quantify the impact of parameter uncertainty on the estimated capacitor lifetime, `main.py` implements a Monte Carlo–based reliability assessment following the methodology proposed in [4]. In the original work, this approach was applied to power semiconductor devices (IGBTs and diodes); in this study, the same framework is adapted and extended to DC-link capacitors. Key parameters, namely the effective ESR ESR_{eff} , insulation resistance R_{ins} , thermal resistance R_{th} , and the thermal calibration factor k , are sampled from normal distributions with a certain standard deviation around their nominal values. Using mean operating stresses (V_{cap} , I_{cap} , T_{env}), the sampled parameters are propagated through the capacitor thermal model to obtain a statistical distribution of the core temperature T_{core} . The resulting temperature samples are then mapped to lifetime using the graph-based lifetime model, yielding a probabilistic lifetime distribution.

Important Notes

- **Temporal resolution of mission profiles:** The mission-profile inputs P , Q , and T_{env} are provided with a resolution of 1 s. All derived electrical quantities such as V_s and V_{dc} are therefore also defined at a 1 s resolution on the system level.
- **Core temperature calculations:** The capacitor core temperature is estimated using a simplified steady-state thermal model of the form

$$T_{\text{core}} = T_{\text{amb}} + P_{\text{loss}} \cdot R_{\text{th}}$$

In principle, more detailed thermal representations such as Foster or Cauer networks could be employed; however, these models require thermal time constants and capacitances that are typically not provided in capacitor datasheets. Consequently, a lumped thermal resistance model is adopted, representing the highest level of thermal fidelity achievable with the available manufacturer data.

- **Multiple lifetime models:** The framework implements multiple capacitor lifetime models to account for differences in capacitor technology, manufacturer design, and dominant ageing mechanisms. Separate models are provided for aluminum electrolytic capacitors and film capacitors along with generic voltage–temperature accelerated ageing formulations which can be used for both aluminum electrolytic capacitors and film capacitors. Since capacitor lifetime is primarily governed by temperature and voltage stress, manufacturers typically propose empirical lifetime models tailored to their specific materials, construction and failure physics.

The lifetime models implemented in this work correspond to the most widely used and representative formulations reported in the literature and manufacturer documentation, as summarized in [3]. In practice, capacitor datasheets most often provide lifetime information in the form of voltage–temperature lifetime graphs. For this reason, the graph-based lifetime interpolation model is recommended and used as the default approach in this framework, as it directly reflects manufacturer-qualified lifetime data without requiring additional empirical assumptions.

4 Case Study

To demonstrate the functionality of the proposed electro–thermal lifetime model of DC link capacitor, a case study was performed. The objective of this study is to evaluate how different power factors influence the lifetime of the DC Link capacitor. A mission profile of constant electrical loading over a 24 hour interval was selected. In practice, the model simulates this 24 hour segment and then extrapolates the resulting thermo–mechanical degradation to estimate how long the switch would survive if it were to operate continuously under these repeated 24 hour batches.

The apparent power was fixed at the rated value of the inverter, while the active and reactive power components were adjusted to generate the desired power factors. The ambient temperature was held constant at 35°C, and both the DC-side voltage and AC-side phase voltage were also kept constant for all simulations. A DC-link metallized polypropylene film capacitor (TDK–EPCOS, model B32778J8806K000, 80 μ F, 800 V DC, $\pm 10\%$ tolerance) was considered in this study [5]. All required electrical, thermal and lifetime-model input parameters for these components have already been detailed in Section 3.1. The full repository contains the exact numerical values used in the simulation, therefore, individual parameter values are not repeated here.

4.1 Results and Discussions

Figure 2 illustrates the inverter RMS phase current as a function of the power factor for a constant apparent power operation. As the apparent power is fixed, the phase current remains constant for different power factor values. This behavior confirms that changes in power factor primarily affect the phase angle rather than the magnitude of the inverter current under constant apparent power conditions.

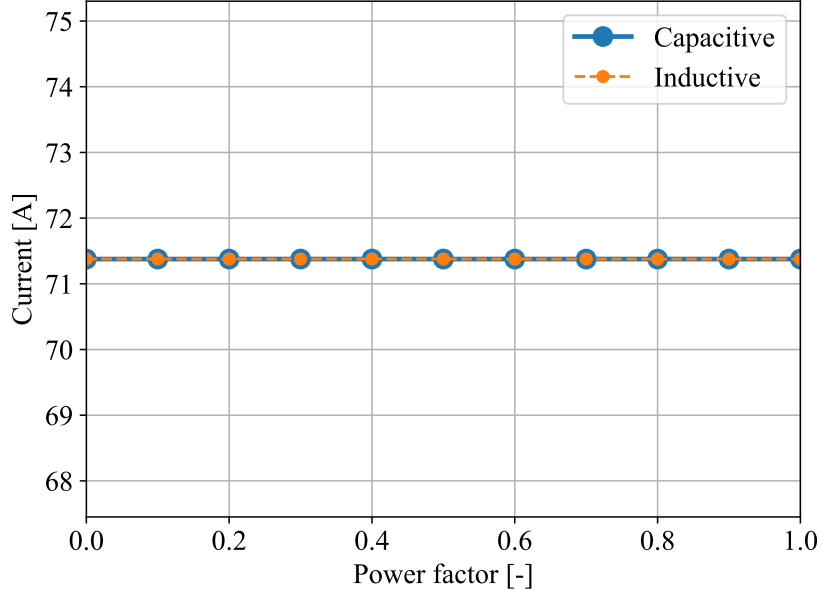


Figure 2: Average inverter RMS phase current.

Figure 3 shows the RMS DC-link capacitor ripple current as a function of the power factor for constant apparent power operation. Unlike the inverter phase current, the DC-link ripple current exhibits a pronounced dependence on the power factor. As the power factor deviates from unity toward both inductive and capacitive operation, the ripple current increases, reaching higher values under predominantly reactive power conditions.

The increase in DC-link capacitor RMS ripple current with decreasing power factor originates from the growing reactive power exchanged between the inverter and the AC side. At unity power factor, most of the inverter current contributes to active power transfer, resulting in relatively smooth DC-link power flow. As the power factor decreases, a larger portion of the phase current becomes reactive, leading to increased instantaneous power oscillations at twice the grid frequency. These oscillations cannot be supplied by the DC source and must be buffered by the DC-link capacitor, thereby increasing the capacitor RMS ripple current. This mechanism has been analytically demonstrated in [1], which shows that the DC-link capacitor current depends strongly on the phase angle between voltage and current, even under constant apparent power operation.

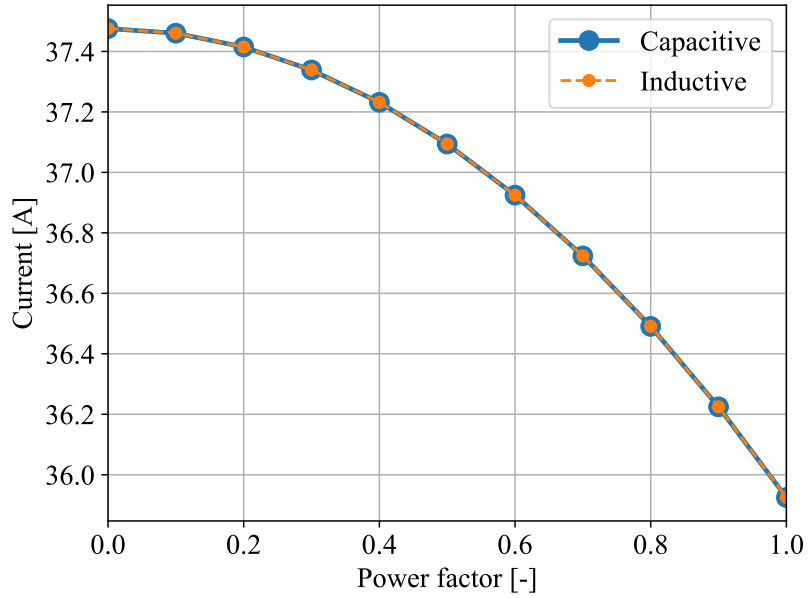


Figure 3: Average DC ripple RMS current.

Figure 4 shows the RMS current flowing through the DC-link capacitor as a function of the power factor. In the considered case study, only a single DC-link capacitor is used in the converter. Consequently, the entire DC-link ripple current is carried by this capacitor, and no current sharing occurs within the DC-link capacitor bank. As a result, the per-capacitor RMS current closely matches the DC-link RMS ripple current shown in Fig. 3. The observed dependence on power factor is therefore identical, confirming that variations in power factor directly translate into changes in the electrical stress experienced by the DC-link capacitor.

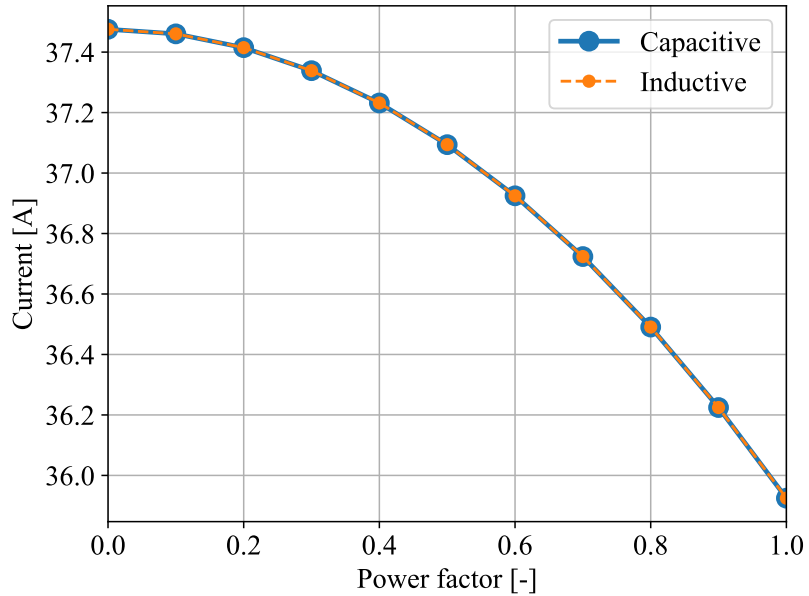


Figure 4: Average DC ripple RMS current per capacitor.

Figure 5 shows the estimated DC-link capacitor core temperature as a function of the power factor under constant apparent power operation. As the power factor deviates from unity toward inductive or capacitive operation, the core temperature increases. This trend directly follows the increase in DC-link RMS ripple current at lower power factors, which leads to higher internal power dissipation due to ripple-current losses in the capacitor ESR. Since the ambient temperature and thermal resistance are kept constant in this case study, variations in the core temperature are solely driven by changes in electrical losses. The results highlight that operation at low power factor significantly increases the thermal stress on the DC-link capacitor, thereby accelerating ageing and reducing lifetime.

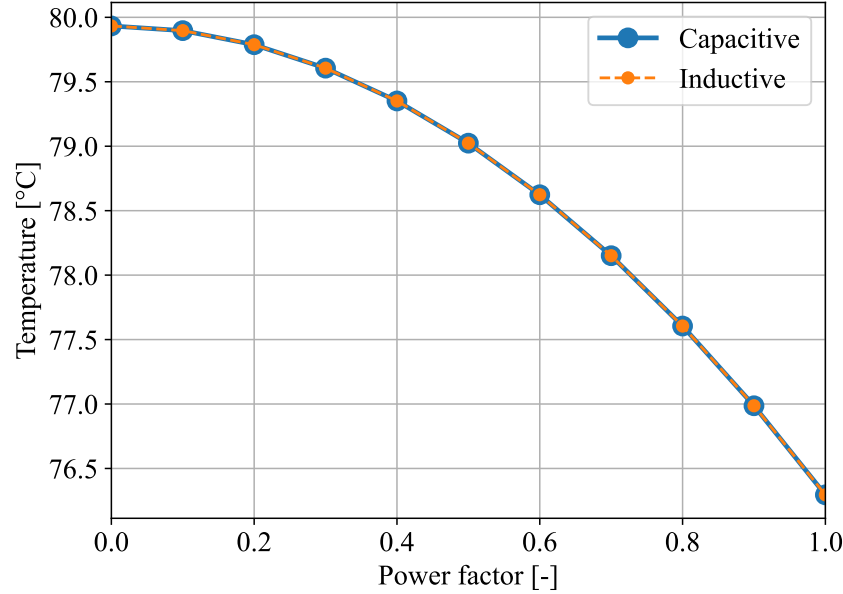


Figure 5: Average DC link capacitor core temperature.

Figure 6 presents the estimated total lifetime of the DC-link capacitor as a function of the power factor for the considered case study. The lifetime decreases as the power factor deviates from unity toward both capacitive and inductive operation. This trend is a direct consequence of the increased DC-link RMS ripple current at lower power factors, which leads to higher internal power dissipation, elevated core temperature and accelerated ageing. Therefore, operation under predominantly reactive power conditions significantly shortens the lifetime, even though the apparent power and ambient temperature remain constant. These results clearly demonstrate that power factor plays a critical role in DC-link capacitor reliability and should be carefully considered in converter operation and design.

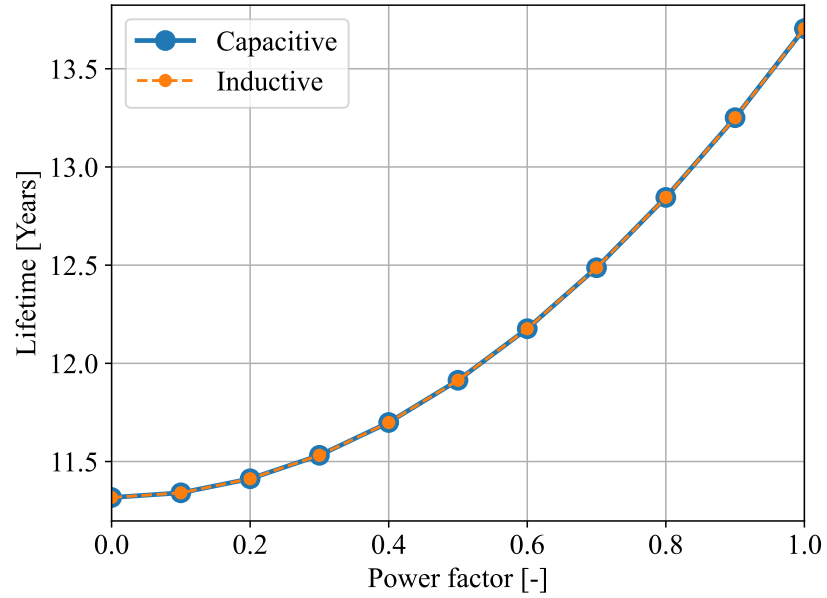


Figure 6: DC link capacitor lifetime.

Overall, the case study highlights that power factor has a significant impact on DC-link capacitor lifetime, even under constant apparent power operation. While the inverter phase current remains unchanged, lower power factor leads to increased DC-link ripple current, higher core temperature, and accelerated ageing of the capacitor. As a result, operation with high reactive power content substantially reduces the expected lifetime, underscoring the importance of considering power factor effects in DC-link capacitor design and reliability assessment.

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